Andhra University College of Engineering

Electronics and Communication Engineering | Course Code: EC301

Digital Signal Processing - Comprehensive Syllabus Lecture Notes

Syllabus Focus: EC301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Digital Signal Processing. Includes detailed

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Discrete Fourier Transform & FFT

" Syllabus Scope: Properties of DFT, Circular convolution, Radix-2 Decimation-In-Time DIT and Decimation-

Unit 2: Infinite Impulse Response IIR Filters

" Syllabus Scope: Analog filter approximations Butterworth, Chebyshev, Bilinear transformation method, Im

EC301: Digital Signal Processing - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Finite Impulse Response FIR Filters

" Linear phase FIR filter characteristics, Symmetric and antisymmetric impulse response, Windowing techniques

Unit 4: Realization of Digital Filters

" Direct Form I, Direct Form II, Cascade, Parallel, and Lattice ladder realizations for IIR and FIR filter arc

Unit 5: Finite Word Length Effects

" Fixed-point and floating-point representations, ADC quantization noise, Coefficient quantization, Round-off

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Compute the 8-point DFT using Radix-2 DIT-FFT butterfly signal flow graph with twiddle factor computat
2. Design an analog Butterworth low-pass filter satisfying specified passband ripple and stopband attenua
3. Formulate the transfer function of a linear phase FIR high-pass filter using the Hamming window techni

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Alan V. Oppenheim, Ronald W. Schafer - Discrete-Time Signal Processing Pearson 3rd Ed
* John G. Proakis, Dimitris G. Manolakis - Digital Signal Processing Pearson 4th Ed

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